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From Differentiable Models to Hybrid Engines: A 2025-2026 Survey of Open-Source Physical Piano Synthesis

Emergence of Differentiable Physical Modeling Frameworks

The period from 2025 to the present has witnessed a significant methodological shift in the domain of physical piano synthesis, marked by the rise of differentiable physical modeling frameworks. This approach represents a confluence of traditional physics-based simulation and modern machine learning, leveraging automatic differentiation to optimize model parameters by minimizing the discrepancy between synthesized audio and target reference signals [3]. The ISMIR 2025 tutorial on "Differentiable Physical Modeling Sound Synthesis" explicitly highlighted this trend as a central theme, underscoring its growing importance within the research community [3]. This methodology moves beyond purely analytical formulations or empirically tuned parameters, instead framing the synthesis problem as an optimization task where the model itself can be refined using large datasets. This allows for the creation of highly complex, non-linear systems that can achieve greater perceptual fidelity than their more rigid counterparts. The underlying principle involves constructing a computational graph of the entire synthesis process, from MIDI input to raw audio output, and then using gradient descent to adjust the model's learnable parameters until the generated sound closely matches a desired acoustic signature.

A seminal example of this paradigm in action is the work by Simionato et al. on Sines, Transients, Noise Neural Modeling (STN-NM) [7][17][20]. This research proposes a novel hybrid synthesizer architecture that decomposes the complex sound of a piano note into three distinct, physically-motivated components: a quasi-harmonic part, a transient part, and a noise component [7]. The quasi-harmonic component, representing the main body of the tone, is modeled using a differentiable sinusoidal model. Crucially, this model is not a black-box neural network; it is guided by explicit physics-derived formulas. It incorporates fundamental principles of string vibration, such as inharmonicity calculated from Timoshenko beam equations, frequency-dependent damping based on the Valette &

Cuesta (VC) model, and phenomena like double decay stages resulting from vertical and horizontal string polarizations [17] [20]. This fusion of a physics-informed structure with learnable parameters provides both interpretability and flexibility. The transient component, responsible for the initial hammer-strike impact, is synthesized using a deep convolutional network designed to generate waveforms transformed via an Inverse Discrete Cosine Transform (IDCT) [20]. The noise component, which captures the sound of the hammer felt striking the string and other high-frequency irregularities, is modeled as filtered white noise, where the filter coefficients themselves are learned by the network [7] [20]. The entire system is implemented as a PyTorch-based project available on GitHub at `https://github.com/RiccardoVib/STN_Neural` [7] [17].

The perceptual quality of the STN-NM model was rigorously evaluated through a Multiple Stimuli test with Hidden Reference and Anchor (MUSHRA) listening test [7] [17] [20]. This formal evaluation involved twenty trained musician participants who rated single notes and trichords generated by the model against a high-quality reference recording from the same piano [17]. The results indicated that while the synthesized sound was perceptually distinguishable from the real piano, the difference was not substantial [7]. A primary area of limitation identified by the participants was inaccuracies in the attack phase of the notes [7] [17]. Despite this, the model demonstrated a notable capability: when generating chords, its output was rated as better than simply summing the individual synthesized notes, suggesting a degree of plausible interaction between partials [20]. Statistical analysis using the Friedman test confirmed that the observed differences were both statistically significant (p-values < 0.05) and practically relevant (large effect sizes), validating the perceptual findings [17] [20]. The model's architecture is also noted for its computational efficiency compared to previous methods, with approximately 180,000 trainable parameters (excluding reverb), making it suitable for low-latency, interactive applications [14] [20]. The dataset used for training consisted of recordings from the BiVib dataset, focusing on two octaves (C3 to B4) from both upright-closed and grand-open pianos, with each recording associated with a velocity range between 30 and 100 [17].

This differentiable approach extends beyond the piano to other distributed musical systems. V Zheleznov's 2025 work examines how modal decomposition can be combined with neural ordinary differential equations (ODEs) for modeling distributed musical systems [15]. While not specific to the piano, this research points to a broader trend of using advanced mathematical constructs, informed by physics, to describe the dynamics of vibrating bodies. Similarly, the concept of Differentiable Modal Synthesis for Planar String Simulation, published in November 2025, demonstrates a computationally

efficient method for modeling complex vibrational modes [12]. An associated MambaSCI repository provides a concrete implementation of this technique [13]. These developments collectively signal a move towards more expressive, optimized, and physically grounded synthesis engines that can be tailored to specific instruments and performance styles through data-driven refinement. The emphasis on providing open-source code alongside these papers, as seen with the STN-NM model, is critical for enabling reproducibility and further innovation in the field [7] [17].

Leading Open-Source Full Instrument Synthesis Implementations

In the pursuit of full instrument physical synthesis for the piano, several open-source projects have emerged from 2025 to the present, offering tangible implementations of complete signal paths. These projects prioritize end-to-end modeling, encompassing the hammer strike, string vibration, and crucially, the subsequent excitation of the soundboard and body resonance. Among these, one project stands out for its comprehensive, albeit early-stage, implementation, providing a valuable baseline for researchers and developers.

The `coopiano` project, available as an open-source synthesizer in C++17, represents a direct and modern attempt to model a complete piano instrument from first principles [19]. Developed under the MIT license, which permits broad reuse including in commercial applications, the project's codebase is designed for accessibility and integration [19]. Its synthesis engine meticulously models the entire chain of acoustic events: it includes components for the hammer, the string, the soundboard, sympathetic resonance, and even a final room reverb effect, creating a holistic simulation of a virtual piano in an acoustic space [19]. The architecture is well-documented, featuring a header-only synthesis library (`piano.hpp`) that can be used independently, as well as specialized components like `streaming_soundboard.hpp` for convolution-based soundboard modeling [19]. This modular design facilitates experimentation with individual components without needing to rebuild the entire engine. The project supports offline rendering of notes and entire MIDI files to stereo WAV format across multiple platforms via a standard Makefile, demonstrating its portability [19]. Furthermore, it provides live MIDI input exclusively on macOS using native system frameworks (CoreMIDI, AudioToolbox), indicating a clear path for interactive use [19]. However, it is important to note the developer's own assessment that the project is in an early release state ("VERY EARLY days"), and features

like AU and VST plugins are not included in this initial open release [19]. This suggests that while the core algorithmic implementation is robust and complete, it may lack the polish, user interface, and feature set of mature commercial products. Nonetheless, as a documented, permissively licensed, end-to-end physical model, `coopiano` serves as an invaluable resource for anyone wishing to study or build upon a comprehensive open-source piano synthesis engine.

Another significant open-source project is the physically modeled piano synthesizer mentioned in one source, which is available under the GNU General Public License v3.0 (GPL-3.0) [2]. This project is notable for its practical application as a plugin synthesizer, providing compiled binaries for major platforms including VST, VST3, AU (macOS), and LV2 (Linux), making it immediately usable within digital audio workstations [2]. The technology is based on physical modeling synthesis using digital waveguide string models, a well-established and computationally efficient method for simulating string vibration [2]. The implementation goes beyond simple strings by including simulations for hammer-string interaction, longitudinal string modes, and soundboard resonance [2]. This combination of features aligns with the goal of achieving full instrument synthesis. The project is built with modern C++ standards (C++17) and requires CMake 3.16 or higher for building, with platform-specific dependencies clearly outlined for macOS, Windows, and Linux [2]. A particularly strong aspect of this project is its commitment to quality assurance and reproducibility. It includes a comprehensive test suite with unit tests for core components like filters, the hammer model, and the digital waveguide string module, as well as integration tests such as scale and chord tests and a lengthy 4-minute stress test [2]. This level of testing provides confidence in the correctness of the individual modules that comprise the larger synthesis engine. While less information is provided about its maintenance status or development roadmap compared to `coopiano`, its availability as a functional plugin with a GPL-3.0 license makes it a powerful tool for users who need a ready-to-use solution and are comfortable with the implications of that license, which requires derivative works to also be open-source.

Finally, the MAESSTRO project, hosted on GitLab at `https://gitlab.com/benjamin.elie/maesstro`, offers a glimpse into the work of influential researchers in the field [1]. Developed by Benjamin Elie, the project is connected to a discussion thread mentioning other key figures such as Antoine Chaigne, Juliette Chabassier, and Balázs Bank, who are all recognized contributors to the theory and practice of physical piano modeling [1]. While the project may not be actively maintained, its value lies in providing access to the architectural choices and implementations of researchers who have shaped the discipline. It serves as a historical artifact and a starting point for understanding established methodologies. The existence of a dedicated discussion thread further

validates its relevance and credibility within the research community [1]. Collectively, these projects—`coopiano` for its foundational completeness, the GPL-licensed synthesizer for its practical plugin utility, and MAESSTRO for its connection to foundational research—provide a rich landscape of open-source options for those interested in full instrument physical synthesis.

Component-Level Innovations in String and Hammer Modeling

While the ultimate goal is often full instrument synthesis, significant advancements in the realism of physical piano models frequently originate at the component level, particularly in the accurate simulation of string vibration and the hammer-string interaction. The period from 2025 to the present has produced several noteworthy open-source contributions and research directions in these areas, moving beyond basic waveguide models to incorporate more sophisticated physical phenomena.

One of the most prominent trends in string modeling is the continued exploration and application of variants of the Karplus-Strong (KS) algorithm. Although originally developed in the late 1970s, its descendants remain relevant due to their simplicity and effectiveness in generating plucked and struck string tones. The Extended Karplus-Strong Synthesizer, known as Soundkist, is an open-source VST plugin that demonstrates the ongoing interest in this family of algorithms [11]. By providing a modern plugin wrapper around the KS principle, it makes these classic synthesis techniques accessible to contemporary musicians and producers [11]. More innovatively, the Resonarium digital waveguide synthesizer integrates a damping filter specifically designed to facilitate live interpolation between a standard Karplus-Strong style of string synthesis and more complex, banded waveguide methods [10]. This approach does not discard KS synthesis but rather uses it as one end of a continuum, allowing for dynamic morphing between timbres of varying complexity. This is a novel strategy that leverages the computational efficiency of KS models while retaining the ability to transition to richer, more resonant sounds, showcasing a flexible approach to string modeling within a single framework.

Beyond KS variants, modal synthesis has emerged as a powerful alternative for modeling complex vibrations. A paper from November 2025 introduced Differentiable Modal Synthesis for Physical Modeling of Planar String Sound and Motion Simulation [12]. This technique focuses on decomposing the string's motion into its constituent vibrational

modes, which can then be simulated individually. This approach can be computationally efficient and offers a natural way to incorporate complex boundary conditions and material properties. The associated research is supported by a GitHub repository named MambaSCI, which contains the open-source implementation of this method [13]. This project provides a concrete tool for researchers interested in exploring modal synthesis from a differentiable perspective. The extension of these ideas to distributed musical systems is also being explored, with research examining the combination of modal decomposition with neural ordinary differential equations (ODEs) [15]. Such work points toward a future where the vibrational modes of a piano soundboard could be modeled with unprecedented physical detail and expressiveness.

The hammer-string interaction is arguably the most critical component for capturing the piano's characteristic percussive attack. Several open-source projects explicitly model this non-linear event. The `coopiano` synthesizer, for instance, includes a dedicated hammer component as part of its complete physical model, ensuring that the initial strike on the string is accurately represented [19]. Similarly, the open-source synthesizer released under the GPL-3.0 license also features a detailed simulation of the hammer-string interaction [2]. One GitHub repository, `lucainiaoge/piano-physics-model`, describes a simulation based largely on the work of Balázs Bank, a prominent researcher in the field whose publications provide foundational insights into hammer-string mechanics [18]. The fact that his work is being referenced and implemented in open-source projects underscores its significance and the community's effort to make these advanced concepts publicly accessible. The simulation described in this repository models a system comprising the string, hammer, bridge, and soundboard, demonstrating a multi-component approach to modeling [18]. Accurately capturing the stiffness of the hammer felt, the non-linear force-distance relationship during the strike, and the subsequent transfer of energy to the string are all essential for a convincing piano sound, and the inclusion of these features in multiple open-source projects highlights their recognized importance.

Hybrid AI and DSP Approaches to Piano Synthesis

Parallel to the advancements in purely physical and differentiable modeling, a distinct and increasingly popular trend involves the hybridization of artificial intelligence with traditional Digital Signal Processing (DSP). This approach seeks to leverage the strengths of both paradigms: the data-driven adaptability and pattern recognition capabilities of neural networks, and the computational efficiency, stability, and physical interpretability

of classical DSP. The period from 2025 to the present has seen the emergence of several open-source projects that exemplify this dual-engine philosophy, aiming to balance high fidelity with the real-time performance required for interactive music production.

A prime example of this hybrid methodology is 'Piano Forte', a free and open-source virtual instrument developed by Omnes Sonos [4]. Piano Forte utilizes a dual-engine architecture that blends a lightweight neural network with a physical model. The neural network component is a relatively simple feedforward type, but its power comes from the sophisticated preprocessing of the training data [4]. The segmentation algorithm used to chop sample sets into pseudo-cycles in the frequency domain is an upgraded version of a technique from Carlos Tarjano's Ph.D. thesis, lending significant academic credibility to the project [4]. The neural network is trained on piano samples that are available under a Creative Commons license. The physical model component complements the neural network, likely handling aspects of the sound that are more amenable to physical simulation, such as the long-term decay and harmonic structure. This architecture is designed to achieve a balance between computational efficiency and high-fidelity audio output, making it suitable for demanding real-time applications [4]. The source code is available on GitHub, and the project is distributed as a plugin in VST3, Standalone Windows (x64), and LV2 formats, ensuring broad compatibility [4]. The developer, Carlos Tarjano, noted that the initial release was an MVP built over about a year, and he has articulated a clear long-term vision: to bridge the quality gap between Piano Forte and the leading commercial product, Pianoteq [4]. This forward-looking statement positions Piano Forte not as a finished product, but as an active and promising research project with significant potential for future development.

Another relevant direction is found in the work of Riccardo Simionato, who, in collaboration with others, has proposed hybrid modeling approaches that combine deep learning with traditional DSP within a differentiable framework [14]. Their research builds upon earlier proposals, including a "Physics-informed differentiable method for piano modeling" [20]. This work focuses on single piano notes and uses a neural network to learn a quasi-harmonic spectrum. The network optimizes parameters derived from physics-based formulas for string inharmonicity and partial envelope decays, effectively guiding the learning process with physical constraints [14]. This approach maintains a modular architecture, operates with low latency, and has a modest computational load, making it suitable for real-time interactive applications [14]. While their most detailed public work, the STN-NM model, focuses on spectral components, the general principle of combining differentiable DSP modules with learnable neural networks is a powerful hybrid strategy. The release of the 'ToneTwist AFx dataset' by Comunitá et al., which consists of dry/wet pairs for various audio effects, further supports this trend by

providing a standardized resource for training and evaluating such hybrid emulation models [14].

These hybrid approaches contrast with purely physical models like `coopiano` [19] and purely data-driven models like Piano-SSM. Piano-SSM, introduced by Dallinger in 2025, is an end-to-end deep diagonal State Space Model (SSM) that synthesizes raw piano audio directly from MIDI input [5] [24]. Its repository is available on GitHub at `https://github.com/domdal/piano-ssm` [23] [25]. While it achieves impressive efficiency and real-time performance, it represents a completely different paradigm from the physics-guided synthesis seen in other projects. The hybrid models like Piano Forte and Simionato's differentiable DSP work occupy a middle ground, seeking to capture the best of both worlds: the perceptual richness that data can provide and the structural integrity and efficiency of physical laws. This makes them particularly compelling for developers aiming to create instruments that are not only realistic-sounding but also responsive and computationally manageable.

Comparative Analysis of Identified Projects and Methodologies

A comparative analysis of the identified open-source physical piano synthesis projects reveals a diverse and rapidly evolving landscape. Each project employs a distinct methodology, prioritizing different trade-offs between perceptual quality, computational efficiency, physical accuracy, and implementation maturity. No single project emerges as definitively superior across all metrics; instead, the choice of tool depends heavily on the user's specific goals, whether they be academic research, artistic creation, or commercial development. The following table provides a structured comparison of the key projects discussed, highlighting their core technologies, licensing, and accessibility.

Project / Paper	Core Technology	Key Features & Innovations	Licensing	Code Availability
Simionato et al. (STN-NM)	Differentiable Spectral Modeling [7]	Decomposes sound into Sines, Transients, Noise. Physics-guided differentiable sinusoidal model. Explicitly models inharmonicity, damping, beatings, and phantom partials.	Not specified in sources	GitHub: `RiccardoVib/STN_Neural` [7] [17]
Dallinger (Piano-SSM)	Deep Diagonal State Space Model (SSM) [5]	End-to-end neural network synthesizing raw audio from MIDI. Designed for high efficiency and real-time performance.	Not specified in sources	GitHub: `domdal/piano-ssm` [23] [25]
`coopiano`	Complete Physical Modeling	End-to-end model of hammer, string, soundboard, sympathetic resonance, and room reverb. Header-only C++ library.	MIT License [19]	GitHub/GitLab (implied by context)
GPL-3.0 Synthesizer	Digital Waveguide Synthesis	Includes hammer-string interaction, longitudinal string modes, and soundboard resonance. Comes with pre-compiled plugins.	GPL-3.0 [2]	Source code provided with project
Omnes Sonos (Piano Forte)	Hybrid AI/DSP	Dual-engine architecture blending a lightweight neural network with a physical model. Uses advanced segmentation from a PhD thesis.	GPL-3.0 [4]	GitHub [4]
MAESSTRO	Legacy Physical Modeling	Open-source project by Benjamin Elie, connecting to foundational research by Chaigne, Bank, and Elie.	Not specified in sources	GitLab: `benjamin.elie/maesstro` [1]
Soundkist	Extended Karplus-Strong Synthesis	Open-source VST plugin based on the Extended Karplus-Strong algorithm.	Not specified in sources	VST Plugin (source not available) [11]

The methodologies can be broadly categorized. At one end of the spectrum is the differentiable modeling approach of the STN-NM project [7] [17]. Its greatest strength lies in its foundation in physics, which ensures that the synthesized sounds adhere to physical laws, while the differentiable components allow for fine-tuning based on perceptual feedback. The rigorous MUSHRA listening test provides a strong basis for its perceptual quality claims, though the identified weakness in the attack phase remains a key area for improvement [7]. This makes it an excellent candidate for research focused on improving specific aspects of piano tone, especially the initial transient.

At the opposite end is the fully data-driven Piano-SSM [5] [24]. This approach sacrifices explicit physical modeling in favor of pure data-driven generation. Its primary advantage is efficiency and ease of use, as it operates as a simple MIDI-to-audio converter. However, it lacks the interpretability and control offered by physics-based models. The trade-off is between high fidelity achieved through learning and high efficiency achieved through abstraction.

The `coopiano` project represents a middle ground of pure physical modeling [19]. It aims for a complete, end-to-end simulation of the piano's acoustic chain. Its permissive MIT license makes it highly attractive for both research and commercial development, and its

modular C++ library structure is ideal for integration into other software [19]. Its main limitation is its early developmental stage, which may mean it lacks the sonic refinement of commercial counterparts [19]. The GPL-3.0 licensed synthesizer offers a similar physical modeling philosophy but with the immediate usability of pre-compiled plugins [2]. The choice between `coopiano` and this GPL project would depend on the user's need for a library vs. a plugin and their willingness to comply with the stricter terms of the GPL license.

Finally, the hybrid AI/DSP approach of Piano Forte represents a pragmatic compromise [4]. By combining a neural network with a physical model, it attempts to get the best of both worlds: the adaptability of AI and the efficiency and stability of DSP. Its reliance on a segmentation algorithm from a PhD thesis gives it academic weight, and its developer's stated ambition to close the quality gap with Pianoteq indicates a promising future trajectory [4]. This makes it a compelling option for users who want a ready-to-use, high-quality instrument that is also backed by a solid research foundation. The MAESSTRO project serves as a historical reference, providing insight into the methodologies of foundational researchers [1].

It is critical to acknowledge a significant gap in the available information: there is no direct, head-to-head perceptual comparison of these open-source projects against each other or against the leading commercial product, Modartt's Pianoteq [8] [9]. Pianoteq is consistently cited as a benchmark for physical modeling realism, sometimes being described as more realistic than even the most meticulously sampled libraries [8]. However, its proprietary algorithms are not publicly dissected in the provided materials. Therefore, any conclusion about the absolute "best" open-source solution must be qualified by the knowledge that a commercial product exists that sets a very high bar for perceptual quality. The identified open-source projects represent the cutting edge of publicly available, reproducible research, but they operate in a competitive environment where a leading closed-source product continues to push the boundaries of what is possible.

Synthesis and Strategic Recommendations for Researchers

The investigation into the best and latest open-source physical piano modeling papers and code from 2025 to the present reveals a vibrant and multifaceted field undergoing

significant evolution. The dominant trends point away from monolithic, purely analytical models and toward more flexible, hybrid, and optimization-driven frameworks. The primary criterion for identifying "best" implementations, as stipulated by the research goal, is perceptual audio quality, followed by measured physical accuracy. Based on the available evidence, several strategic recommendations can be made for researchers and developers seeking to engage with this domain.

First and foremost, the most innovative and promising direction for high-fidelity, perceptually-oriented synthesis is the adoption of differentiable physical modeling frameworks. The work by Simionato et al. on Sines, Transients, Noise Neural Modeling (STN-NM) stands out as a premier example of this approach [7] [17] [20]. Its methodology of decomposing the piano sound into physically-motivated components—a differentiable sinusoidal model for the quasi-harmonic content, a deep convolutional network for transients, and a learnable filter for noise—is both powerful and interpretable [7]. The explicit incorporation of physical principles like Timoshenko beam theory for inharmonicity and the VC model for damping grounds the model in reality while allowing learnable parameters to capture subtle nuances not accounted for by the physics alone [17] [20]. The provision of open-source code on GitHub and the conduct of a rigorous MUSHRA listening test provide a solid foundation for further research and experimentation [7] [17]. Researchers should prioritize exploring this `STN_Neural` framework as a starting point, as it directly addresses the user's goal of balancing perceptual quality with physical accuracy.

Second, for those pursuing a more traditional, end-to-end physical modeling approach, the `coopiano` project offers an unparalleled resource [19]. As a complete, permissively licensed (MIT) C++17 implementation of a full piano instrument—including hammer, string, soundboard, and room reverb—it provides a robust and extensible baseline [19]. While it is in an early release state, its documented architecture and modularity make it an ideal platform for studying the interactions between different physical components and for implementing new modeling techniques. It serves as a practical realization of the kind of comprehensive models that form the bedrock of the field, and its accessibility accelerates the pace of development and validation.

Third, the growing trend of hybrid AI and DSP architectures, exemplified by the Omnes Sonos 'Piano Forte' project, presents a highly pragmatic and commercially relevant path forward [4]. By intelligently combining a lightweight neural network with a traditional physical model, Piano Forte successfully balances computational efficiency with high perceptual fidelity, a critical consideration for real-time applications [4]. The project's academic pedigree, stemming from a PhD thesis, lends it credibility, and the developer's

clear roadmap for improvement signals that it is an active and evolving piece of technology [4]. This approach is particularly recommended for developers aiming to create usable, high-quality virtual instruments for musicians, as it focuses on delivering a polished and efficient product.

Given the absence of direct, systematic comparisons between these leading open-source projects, a critical next step for any serious researcher would be to conduct their own comparative study. This would involve acquiring the source code for the `STN_Neural` model [7], the `coopiano` synthesizer [19], and the Piano Forte plugin [4]. The researcher could then generate a standardized set of audio test clips, including scales, chords, and single notes played at various velocities. These clips could then be subjected to a formal perceptual listening test, ideally modeled after the MUSHRA protocol used by Simionato et al. [7], involving a panel of trained musicians to evaluate the outputs against each other and a high-quality reference recording. Such a study would provide invaluable, empirical data on the relative strengths and weaknesses of each methodology, helping to clarify which approaches yield the best perceptual results for specific tasks.

Finally, it is imperative to maintain a balanced perspective by acknowledging the existence of the leading commercial product, Pianoteq [8] [9]. The provided sources offer no technical details about its proprietary algorithms, leaving a significant knowledge gap regarding the exact methods used by the industry leader [9]. Any conclusions drawn from the open-source landscape must be carefully qualified by the fact that a closed-source product continues to set a very high standard for perceptual realism. The identified open-source projects represent the cutting edge of publicly accessible, reproducible science, but they coexist in a competitive ecosystem where a proprietary solution pushes the boundaries of what is achievable.

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